Paper review · Visionary

Confidence calibration of fine-tuned OpenVLA

OpenVLA, Kim, Pertsch, Karamcheti et al., 2024 (arXiv:2406.09246).

Pavel Bushuyeu · October 2026

Does fine-tuning change calibration?

Monitors depend on one fine-tuned model

Help-request monitors read token confidence directly (Zollo and Zemel), train a classifier on it (INSIGHT), conformalize it (ReconVLA), learn a value estimate (TD calibration) or fit a calibrator (VLAConf).

Each is fitted to the rollouts of one fine-tuned model.

The gap

I found no controlled comparison of full fine-tuning and LoRA, or of data budgets, for OpenVLA's success calibration.

The practical question: after a new fine-tune, does the old calibrator still work, and how many new labels are needed?

Hypotheses

H1 With a fixed number of training updates, 10% of the demonstrations gives a larger confidence-minus-success gap than 100%. At matched success, LoRA's gap is larger than full fine-tuning's.

H2 A calibrator fitted on one fine-tune and reused on another predicts success worse than one refitted with the same number of new labels.

H3 For a fixed acting adapter, an ensemble and Laplace-LoRA, tested separately, beat the best simple calibrator on held-out NLL.

H4 Under a mild LIBERO-Plus shift, LoRA's gap grows more than full fine-tuning's, with calibrators frozen from unshifted data.

If H1 and H2 hold, every new fine-tune needs new labeled rollouts, and H2's learning curve says how many. If H3 holds, the adapter brings its own repair.

Design

Stages	1: rank-32 LoRA on 10% and 100% of one suite, three training replicates. 2: full fine-tuning. 3: Bayesian LoRA, then the shift.
Training	A fixed number of updates per run; nested subsets sampled per task.
Confidence	First-step score; success prior, Platt, action-wise Platt; ensemble; Laplace-LoRA.
Evaluation	500 test rollouts per checkpoint; separate validation start states; logits saved.
Metrics	Held-out NLL and the signed gap first; Brier, ECE, AUROC, failures caught.
Splits	Task-balanced 20% / 80%; a learning curve over calibration size; paired comparisons.

Compute and risks

Compute (estimate)

Stage 1: 6 LoRA runs, about 60–90 A100-hours, plus about 30 GPU-hours of evaluation. One run is timed first.

Stage 2: 6 full fine-tuning runs, 240–720 A100-hours, only with a confirmed allocation.

Risks

Token confidence is a proxy, weakest late in failed episodes.

LIBERO-Plus can collapse OpenVLA (1.1% success under camera, 4.4% under lighting changes), so the shift must be mild.

With about 75 failures in 500 episodes, AUROC differences under 0.1 are not detectable.

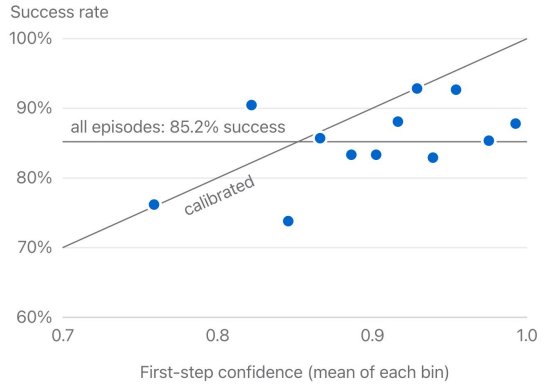
Slightly overconfident, weak on failures

85.2%
success, paper 84.7%

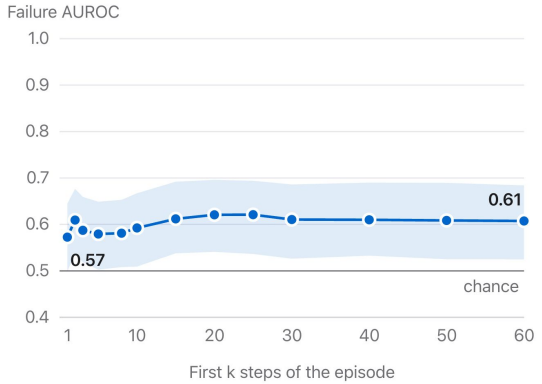
0.061
ECE (study 0.088)

+4.7 pts
overconfident

0.57
failure AUROC



Success stays near 85% at every confidence level.



Over the first 60 steps the AUROC stays near 0.6.

A constant predictor sets the bar

Predictor	Held-out NLL	Brier	Failure AUROC
Raw first-step confidence	0.449	0.128	0.57
Constant success rate	0.425	0.128	0.50
Platt scaling	0.429	0.128	0.54
Action-wise Platt	0.468	0.136	0.52

1,000 task-balanced 20% / 80% calibration and test splits. With about 15 failures in a calibration set, Platt's slope sometimes flips, and action-wise Platt overfits. A threshold that flags 10% of episodes catches 16% of failures. Any new score must beat a constant.

Application: rehearsal in a digital twin

1. **Scan** the workspace and its objects with 1–3 minutes of phone video.
2. **Rebuild** the scene with object meshes, the robot and a matched camera. Set masses and friction by hand, and randomize them.
3. **Rehearse** each candidate checkpoint 200–300 times.
4. **Calibrate** the confidence and set the help threshold.
5. **Deploy**, and compare the failures caught with the prediction of the twin.

What a user gets, and the gaps

Before the first real trial

- Which checkpoint to run in this workspace.
- The failures to expect, as rollout videos.
- The help threshold.

Prior work

SIMPLER and ReVeal rank policies in simulated and reconstructed scenes as real ones do. RialTo builds twins from quick scans.

ReVeal does not recover mass or friction. RialTo needs a person to set objects and physics, and it uses point clouds.

Pilot first, then the transfer test

Pilot (smoke test)

One arm, 2–3 rigid-object tasks, 3 checkpoints × 30 trials in the twin and on the robot.

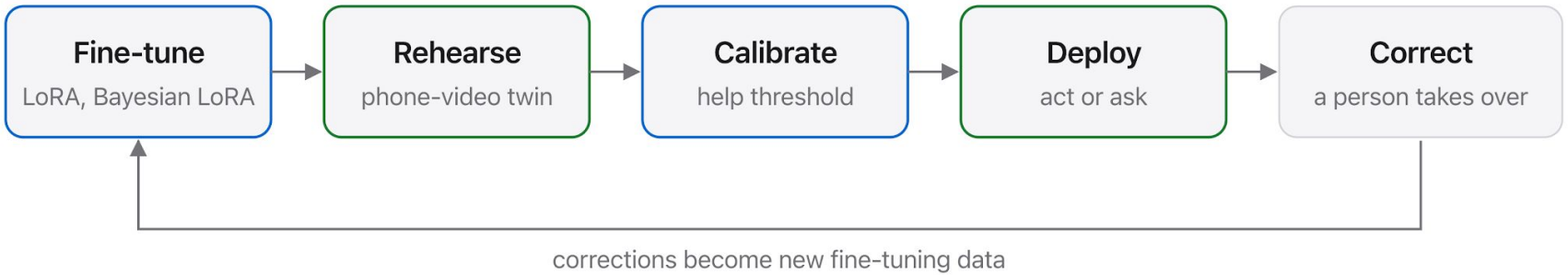
30 trials pin a success rate only to ±18 points, so the pilot checks the pipeline, not the transfer.

Transfer test

The twin calibrator and threshold are frozen, then scored on separate real rollouts against real-only and twin-plus-real calibrators with the same number of real labels.

Held-out NLL, the signed gap and failures caught at a fixed help rate, judged by intervals.

How the two parts connect



■ research project: fine-tuning and calibration

■ application: twin and deployment

What this builds on

Work	What it does	What this adds
Zollo and Zemel, 2025	Calibration of released VLAs on LIBERO.	Varied fine-tuning.
TD calibration, 2026	Episode-level calibration by value estimates.	Varied fine-tuning.
INSIGHT, ReconVLA, VLAConf	Help triggers, conformal or learned confidence.	Twin rollouts.
Laplace-LoRA, ensembles, Römer et al.	Bayesian or ensemble uncertainty.	Token-based OpenVLA.
SIMPLER, ReVeal, RialTo	Evaluation in simulation and twins.	Calibration per workspace.

Sources

1. Kim et al. OpenVLA, arXiv:2406.09246: §5.2–5.3, Table 1 and footnote 4, Appendix E (LIBERO).
2. Zollo and Zemel. Confidence Calibration in VLA Models, 2025.
3. Francis-Meretzki et al. Temporal Difference Calibration, 2026.
4. Karli et al. INSIGHT, 2025. Karli and Fitzgerald. RECALL, 2026.
5. Chen et al. ReconVLA, 2026. Huang et al. VLAConf, 2026.
6. Yang et al. Laplace-LoRA, 2023. Wang et al. LoRA ensembles, 2023.
7. Römer et al. Uncertainty Quantification for Flow-Based Generalist Robot Policies, 2026.
8. Liu et al. LIBERO, 2023. Fei et al. LIBERO-Plus, 2025.
9. Li et al. SIMPLER, 2024. Wang et al. ReVeal, 2026. Torne et al. RialTo, 2024.